

LANGOE, Denmark

WMO: 061380

Lat: 54.821N

Lon: 10.994E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.5	-3.9	-8.6	1.8	-4.5	-6.8	2.1	-3.1	26.4	5.5	23.1	5.2	6.6	120	0.771

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	25.6	19.3	23.8	18.9	22.3	18.3	20.6	23.9	19.6	22.5	18.8	21.5	5.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.1	22.3	18.5	13.3	21.3	17.6	12.6	20.4	59.3	24.0	56.0	22.5	53.3	21.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	15.2	13.4	DB	-6.5	28.5	2.1	2.2	-8.0	30.0	-9.2	31.3	-10.4	32.5	-11.9	34.1	(4)
(5)			WB	-7.3	22.1	1.9	1.6	-8.7	23.2	-9.8	24.1	-10.9	25.0	-12.3	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	1.9	1.6	3.9	8.0	12.4	15.7	18.4	18.4	15.3	10.9	6.5	3.5	(6)
(7)		DBStd	6.63	3.08	3.05	2.77	2.85	2.75	2.37	2.69	2.26	2.20	2.55	2.73	3.17	(7)
(8)		HDD10.0	1086	251	234	189	73	8	0	0	0	0	21	108	201	(8)
(9)		HDD18.3	3204	509	468	447	309	185	85	33	27	95	232	355	459	(9)
(10)		CDD10.0	997	0	0	0	14	82	171	261	261	158	48	3	0	(10)
(11)		CDD18.3	74	0	0	0	0	1	6	36	29	3	0	0	0	(11)
(12)		CDH23.3	202	0	0	0	0	3	14	109	72	4	0	0	0	(12)
(13)		CDH26.7	24	0	0	0	0	0	2	14	9	0	0	0	0	(13)
(14)	Wind	WSAvg	7.2	8.2	7.7	7.3	6.7	6.6	6.6	6.1	6.2	6.8	7.4	7.9	8.5	(14)
(15)	Precipitation	PrecAvg	576	48	37	38	36	38	53	53	63	55	54	52	49	(15)
(16)		PrecMax	715	103	84	88	84	70	113	109	135	131	107	127	97	(16)
(17)		PrecMin	414	0	3	7	15	12	1	12	16	11	18	10	22	(17)
(18)		PrecStd	93	27	18	21	16	15	30	28	35	40	22	29	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.2	12.0	19.2	22.9	25.5	28.6	27.7	23.7	17.6	13.4	9.9	(19)
(20)			MCWB	7.9	6.7	9.1	13.4	16.7	19.8	20.4	18.9	17.6	15.4	12.5	9.1	(20)
(21)		2%	DB	7.5	7.0	9.5	15.8	19.9	22.6	26.1	25.3	21.3	16.0	12.1	8.9	(21)
(22)			MCWB	6.5	6.0	7.6	11.6	15.4	18.3	19.6	18.8	17.1	14.3	11.4	8.1	(22)
(23)		5%	DB	6.7	6.1	8.0	13.7	18.0	20.8	24.3	23.3	19.6	15.0	10.9	8.1	(23)
(24)			MCWB	5.9	5.3	6.6	10.6	14.2	17.3	19.4	18.6	16.5	13.7	10.0	7.5	(24)
(25)		10%	DB	5.8	5.4	7.1	12.1	16.4	19.4	22.8	21.8	18.3	14.2	10.1	7.3	(25)
(26)			MCWB	5.2	4.8	6.0	9.7	13.4	16.3	18.7	18.2	15.9	13.0	9.2	6.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.0	9.3	14.2	18.0	20.8	22.2	21.4	19.2	16.0	12.7	9.5	(27)
(28)			MCD B	8.7	7.7	11.1	17.5	21.7	24.6	25.4	24.5	21.9	17.1	13.2	9.8	(28)
(29)		2%	WB	6.7	6.2	7.9	12.3	16.1	18.9	21.1	20.0	17.8	14.7	11.5	8.3	(29)
(30)			MCD B	7.2	6.8	9.3	15.0	18.7	21.6	24.5	22.8	19.9	15.6	12.0	8.7	(30)
(31)		5%	WB	6.0	5.5	6.9	11.0	14.8	17.8	20.1	19.2	17.0	13.9	10.2	7.6	(31)
(32)			MCD B	6.5	6.1	7.7	13.1	17.1	20.1	23.2	22.0	19.1	14.8	10.8	8.0	(32)
(33)		10%	WB	5.2	4.8	6.2	10.0	13.8	16.7	19.1	18.6	16.2	13.1	9.2	6.8	(33)
(34)			MCD B	5.8	5.4	6.9	11.8	15.9	19.0	22.1	21.2	18.1	14.0	9.9	7.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.4	2.4	3.3	4.7	5.0	5.0	5.3	4.7	3.8	3.0	2.4	2.4	(35)
(36)			MCDBR	2.6	2.8	4.9	7.6	7.6	7.7	7.9	7.4	6.1	3.8	2.8	2.4	(36)
(37)			MCWBR	2.6	2.4	3.5	4.4	4.7	4.8	3.7	3.1	3.3	2.9	2.8	2.5	(37)
(38)		5% WB	MCDBR	2.6	2.8	4.5	7.0	6.6	7.1	7.0	6.0	5.4	3.6	2.7	2.4	(38)
(39)			MCWBR	2.6	2.5	3.5	4.4	4.5	4.9	3.7	3.4	3.4	3.0	2.7	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.341	0.369	0.404	0.396	0.391	0.411	0.411	0.383	0.360	0.339	0.310	(40)	
(41)		taud	2.295	2.314	2.272	2.225	2.337	2.379	2.341	2.350	2.407	2.417	2.351	2.281	(41)	
(42)		Ebn at Noon	607	733	795	816	849	856	829	804	778	706	581	531	(42)	
(43)		Edh at Noon	57	79	102	123	116	113	115	108	90	71	55	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.47	1.09	2.44	4.14	5.29	5.64	5.25	4.21	3.03	1.56	0.62	0.34	(44)	
(45)		RadStd	0.07	0.18	0.23	0.50	0.56	0.38	0.59	0.29	0.30	0.20	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	N/A

Nomenclature: See separate page